Multirate Stein Variational Gradient Descent for Efficient Bayesian Sampling

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Abstract

Many particle-based Bayesian inference methods use a single global step size for all parts of the update. In Stein variational gradient descent (SVGD), however, each update combines two qualitatively different effects: attraction toward high-posterior regions and repulsion that preserves particle diversity. These effects can evolve at different rates, especially in high-dimensional, anisotropic, or hierarchical posteriors, so one step size can be unstable in some regions and inefficient in others. In response, we derive a multirate version of SVGD that updates these components on different time scales. The framework yields practical algorithms, including a symmetric split method, a fixed multirate method (MR-SVGD), and an adaptive multirate method (Adapt-MR-SVGD) with local error control. We evaluate the methods in a broad and rigorous benchmark suite covering six problem families: a 50D Gaussian target, multiple 2D synthetic targets, UCI Bayesian logistic regression, multimodal Gaussian mixtures, Bayesian neural networks, and large-scale hierarchical logistic regression. Evaluation combines distributional fidelity, predictive performance, calibration, mixing, and explicit computational cost accounting. Across this diverse test bed, multirate SVGD variants show improved robustness and more favorable quality-cost tradeoffs than vanilla SVGD, with the strongest gains on difficult hierarchical and strongly anisotropic targets.

Keywords: Stein variational gradient descent, multirate integration, particle methods, kernel Stein discrepancy

1. Introduction

Modern Bayesian inference often requires efficient sampling from complex, high-dimensional, and multimodal posterior distributions. As these problems grow in scale and complexity, traditional sampling methods face significant computational and statistical challenges. The advent of automatic differentiation has enabled the widespread use of gradient-based samplers, making methods such as stochastic gradient Langevin dynamics (SGLD) and stochastic gradient Hamiltonian Monte Carlo (SGHMC) practical and popular choices for large-scale applications [1, 2]. While these approaches advance a single Markov chain using noisy gradients, an alternative paradigm is offered by interacting particle methods, which evolve a population of particles whose empirical distribution approximates the target posterior. This population-based perspective opens new avenues for improving sampling efficiency and robustness, especially in challenging inference settings.

A useful unifying perspective is to view many such methods as discretizations

of a particle flow (possibly with a stochastic component) that transports an initial set of particles toward the target distribution. In this work, we focus on Stein variational gradient descent (SVGD) [3] as a representative interacting particle method. SVGD constructs a velocity field for the particles based on the target density and a reproducing kernel Hilbert space (RKHS) structure, and updates the particles by discretizing this flow. The explicit form of the SVGD update and its velocity field are given in eqs. (1) and (2) in Section 2. The velocity field naturally separates into an attractive drift (moving particles toward high-density regions) and a repulsive interaction (spreading particles to maintain diversity and support coverage of multiple modes).

Despite its conceptual simplicity, SVGD presents a numerical challenge that is central to this paper. The drift and repulsion components can impose different stability constraints on the discretization: when particles crowd, repulsion can become stiff and destabilize updates, while on anisotropic targets the drift can require finer resolution to faithfully follow the geometry of $\log p$. A single step size must therefore balance competing requirements and can either over-damp repulsion (leading to loss of diversity and mode coverage) or under-resolve drift (leading to slow progress). Moreover, repulsion typically requires $O(N^2)$ kernel evaluation per step, so increasing stability by oversampling repulsion can become computationally expensive.

A common approach to solving systems with significantly different components is to use partitioned methods such as implicit-explicit (IMEX) [4, 5] and multirate integration methods [6]. In the context of ordinary differential equations (ODEs), multirate methods exploit a natural splitting of the dynamics into fast and slow components, allowing each to be integrated with a step size appropriate to its scale [7, 8, 9]. For example, in a split ODE of the form $\dot{x} = \phi_{\text{fast}}(x) + \phi_{\text{slow}}(x)$, multirate schemes apply more frequent, smaller steps to the stiff or rapidly varying part, while updating the slow part less often. This approach improves both stability and efficiency in multi-scale dynamical systems, and has been successfully applied in a variety of scientific computing contexts.

The structure of the SVGD update naturally separates into two forces: an attractive drift and a repulsive interaction. These forces often operate on different time scales and can impose distinct numerical challenges. Motivated by advances in multirate time-integration for multi-scale dynamical systems—including MrGARK, IMEX, and related splitting schemes [7, 8, 9, 5, 10]—we propose to integrate the drift and repulsion terms on separate time scales. This leads to a family of multirate SVGD variants, including symmetric splitting, fixed repulsion substepping, and adaptive error-controlled drift substepping.

To assess the effectiveness of these methods, we conduct a comprehensive empirical evaluation. We compare our multirate SVGD variants with established approaches from the literature across a range of benchmark problems, using a variety of task-appropriate metrics. Results highlight the strengths and limitations of each method under diverse geometric and statistical challenges.

1.1. Contributions

Our main contributions are:

- Multirate SVGD formulations that explicitly separate drift and repulsion and support symmetric splitting and fixed multirate schedules.
- An adaptive error-controlled multirate SVGD variant that selects drift substeps based on a local error estimator while keeping repulsion stable.
- A benchmark suite spanning 50D, 2D, UCI, mixture, and BNN targets, with diagnostics (KSD, ESS, mode coverage) and KSD-based early stopping, and with gradient/kernel evaluation accounting for efficiency comparisons.

Organization.. Section 2 presents the SVGD variants and the multirate construction. Section 3 describes the benchmarks and experimental protocol and reports results. Section 4 summarizes findings and discusses directions for future work.

2. Method

2.1. Background: SVGD as a particle flow

Let $p(x)$ denote a target density on $\mathbb{R}^d$ and let $\{x_i^t\}_{i=1}^N$ be a set of particles at iteration t. Stein variational gradient descent (SVGD) evolves particles by following a deterministic interacting-particle flow whose velocity field is constrained to a reproducing kernel Hilbert space (RKHS) [3]. A first-order discretization of this flow is the update

$$x_i^{t+1} = x_i^t + h\,\phi(x_i^t), \tag{1}$$

where $h > 0$ is a step size and the SVGD velocity field is

$$\phi(x) = \frac{1}{N}\sum_{j=1}^{N}\Big[k(x_j, x)\,\nabla_{x_j}\log p(x_j) + \nabla_{x_j}k(x_j, x)\Big]. \tag{2}$$

The first term in eq. (2) is an *attractive* drift that pushes particles toward high-density regions via the score $\nabla \log p$. The second term is a *repulsive* interaction that discourages particle collapse.

Throughout this paper, we limit our focus to an RBF kernel

$$k(x, y) = \exp\Big(-\frac{\|x - y\|_2^2}{2\,h_k}\Big), \tag{3}$$

where the bandwidth h_k.[1]

2.2. Drift–repulsion splitting and multirate integration

The central numerical observation motivating this work is that the drift and repulsion components of eq. (2) can impose different stability constraints

[1] The bandwidth h_k is set by the median heuristic (the median pairwise squared distance among particles) and scaled by a factor $\alpha > 0$ to modulate the strength of repulsion.

and can naturally evolve on different time scales. We therefore write the SVGD flow as a split system

$$\dot{x} = f_{\text{rep}}(x) + f_{\text{drift}}(x), \tag{4}$$

where, for particle i,

$$f_{\text{drift}}(x_i) := \frac{1}{N} \sum_{j=1}^{N} k(x_j, x_i) \nabla_{x_j} \log p(x_j), \tag{5}$$

$$f_{\text{rep}}(x_i) := \frac{1}{N} \sum_{j=1}^{N} \nabla_{x_j} k(x_j, x_i). \tag{6}$$

We next define multirate discretizations that apply different substepping strategies to f_{drift} and f_{rep} while keeping a common macro-step size h.

Strang-split SVGD. We can define the forward Euler steps that advances the split system forward by a stepsize τ as:

$$\Psi^{\tau}_{\text{rep}}(x) = x + \tau f_{\text{rep}}(x), \qquad \Psi^{\tau}_{\text{drift}}(x) = x + \tau f_{\text{drift}}(x).$$

Strang splitting [11] is a classical second order method taht construct a macro-step of size h by composing such substeps. In our setting, one macro step takes the form

$$x^{t+1} = \Psi^{h/2}_{\text{rep}} \circ \Psi^{h}_{\text{drift}} \circ \Psi^{h/2}_{\text{rep}}(x^t), \tag{7}$$

which alternates half a repulsion step, a full drift step, and another half repulsion step. In practice the half-steps can in fact be implemented as $\frac{h}{2m}$ smaller substeps for stability.

Fixed multirate SVGD (MR-SVGD). Fixed multirate integrators provide a principled way to allocate different step sizes to fast and slow components of a split system [6, 7, 8]. In our SVGD setting, however, higher-order multirate schemes are often not cost-effective: their additional stages translate into multiple evaluations of the drift and repulsion fields per macro step, and each such evaluation is expensive due to gradient computations and, in particular, $O(N^2)$ kernel interactions. Moreover, practical accuracy is also limited by finite-particle effects and by choices such as bandwidth selection and early stopping, which can mute the benefits of increasing time-integration order at a fixed computational budget. We therefore specialize to a first-order

Algorithm 1 Fixed multirate SVGD (MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h, repulsion substeps m, kernel k

1: $x \leftarrow \{x_i\}$
2: **for** $s = 1$ to m **do**
3: compute repulsion $f_{\text{rep}}(x)$ using eq. (2)
4: $x \leftarrow x + (h/m)\, f_{\text{rep}}(x)$
5: **end for**
6: compute drift $f_{\text{drift}}(x)$ using eq. (2)
7: $x \leftarrow x + h\, f_{\text{drift}}(x)$
8: **return** x

multirate construction (Euler update maps) that preserves a macro step of size h for the drift while resolving the repulsion with m substeps:

$$x^{t+1} = \Psi^{h}_{\text{drift}} \circ \left(\Psi^{h/m}_{\text{rep}}\right)^{m}(x^t). \tag{8}$$

This retains a coarse drift step while stabilizing the repulsive interaction when particles crowd, at the cost of additional kernel evaluations. The corresponding implementation-level procedure is summarized in algorithm 1.

Adaptive multirate SVGD (Adapt-MR-SVGD). Although fixed-m schedules are simple and robust, they impose a uniform drift resolution along the entire trajectory. As a result, they may spend unnecessary computational effort in benign regions while providing insufficient resolution in locally stiff regimes. To address this imbalance, we choose the number of drift microsteps m adaptively at each macro step, while retaining one repulsion update per macro step. This follows the standard local-error-control perspective used in ODE integrators, where local error estimates are used to adjust resolution [12]. After the repulsion update, denote the state by $x = \Psi^{h}_{\text{rep}}(x^t)$. We estimate the local drift-discretization error by comparing one drift step of size h with two consecutive drift half-steps of size $h/2$,

$$x_{\text{full}} = x + h\, f_{\text{drift}}(x), \tag{9}$$

$$x_{\text{half}} = x + \tfrac{h}{2}\, f_{\text{drift}}(x), \tag{10}$$

$$\tilde{x}_{\text{full}} = x_{\text{half}} + \tfrac{h}{2}\, f_{\text{drift}}(x_{\text{half}}). \tag{11}$$

We then form the relative discrepancy

$$\rho = \frac{\|\tilde{x}_{\text{full}} - x_{\text{full}}\|_2}{\|x\|_2 + \epsilon}. \tag{12}$$

which serves as a dimensionless local error indicator for the drift update. For a first-order drift integrator, the step-doubling discrepancy behaves locally as

$$\rho(h) = Ch^2 + \mathcal{O}(h^3).$$

If the drift step is replaced by m microsteps of size h/m, then

$$\rho_m \approx C(h/m)^2 \approx \rho/m^2.$$

Imposing the target local accuracy $\rho_m \approx \varepsilon$ gives $m \approx \sqrt{\rho/\varepsilon}$. We therefore use the clipped integer controller

$$m \leftarrow \operatorname{clip}\Big(\Big\lceil \sqrt{\rho/\varepsilon} \Big\rceil, m_{\min}, m_{\max}\Big), \tag{13}$$

which increases m when $\rho > \varepsilon$ and decreases it when $\rho \leq \varepsilon$, while enforcing the robustness bounds $m_{\min} \leq m \leq m_{\max}$. The macro-step update is then

$$x^{t+1} = \Big(\Psi_{\text{drift}}^{h/m}\Big)^m \circ \Psi_{\text{rep}}^h(x^t), \qquad m = m(x^t, h). \tag{14}$$

Here $\varepsilon > 0$ is a user-set tolerance and $\epsilon > 0$ is a small constant to avoid division by zero. This strategy increases drift resolution only when the local error indicator suggests stiffness or rapid variation in the drift component. See algorithm 2 for complete implementation.

Remark 1. In implementation, if $m \leq 2$ we reuse the already computed step-doubling trajectory to avoid extra drift evaluations.

Numerical guards. To improve robustness, we clip gradients and clamp extreme state values before and after the split updates, and we replace non-finite updates with the last finite state. These safeguards are implementation-level stability controls and do not alter the intended fixed point of the discretization.

3. Experiments and Results

We evaluate multirate SVGD variants on six benchmark groups: a 50D Gaussian target, a suite of 2D synthetic targets, UCI logistic regression, a 2D Gaussian mixture (mix8), a one-hidden-layer Bayesian neural network (BNN), and a large-scale hierarchical logistic-regression (HLR) benchmark. The comparison focuses on both statistical quality and computational cost.

Algorithm 2 Adaptive multirate SVGD (Adapt-MR-SVGD)

Require: Particles $\{x_i\}_{i=1}^N$, step size h, bounds $m_{\min}, m_{\max}$, tolerance ε

$x \leftarrow \{x_i\}$
compute repulsion $f_{\text{rep}}(x)$ using eq. (6)
$x \leftarrow x + h\, f_{\text{rep}}(x)$
compute drift $f_{\text{drift}}(x)$ using eq. (5)
$x_{\text{full}} \leftarrow x + h\, f_{\text{drift}}(x)$
$x_{\text{half}} \leftarrow x + (h/2)\, f_{\text{drift}}(x)$
compute drift $f_{\text{drift}}(x_{\text{half}})$
$\tilde{x}_{\text{full}} \leftarrow x_{\text{half}} + (h/2)\, f_{\text{drift}}(x_{\text{half}})$
estimate relative error $\rho = \|\tilde{x}_{\text{full}} - x_{\text{full}}\| / (\|x\| + \epsilon)$
$m \leftarrow \text{clip}(\lceil \sqrt{\rho/\varepsilon} \rceil, m_{\min}, m_{\max})$
if $m \leq 2$ **then**
    $x \leftarrow \tilde{x}_{\text{full}}$ ▷ reuse step-doubling path
else
    for $s = 1$ to m **do**
        compute drift $f_{\text{drift}}(x)$
        $x \leftarrow x + (h/m)\, f_{\text{drift}}(x)$
    end for
end if
return x

3.1. Evaluation Metrics

Let $X = \{x_i\}_{i=1}^N$ denote the checkpoint sample set (particles for particle methods; a rolling window for single-chain methods). We use complementary diagnostics that separately assess distributional fidelity, predictive quality, and computational cost.

Moment fidelity. When reference moments are available, we report $e_\mu = \|\hat{\mu} - \mu_{\text{ref}}\|_2$ and $e_\Sigma = \|\hat{\Sigma} - \Sigma_{\text{ref}}\|_F$, where $\hat{\mu} = \frac{1}{N}\sum_{i=1}^N x_i$ and $\hat{\Sigma} = \frac{1}{N}\sum_{i=1}^N (x_i - \hat{\mu})(x_i - \hat{\mu})^\top$. These summarize first- and second-order bias, which is critical in the Gaussian and synthetic-target settings.

Stein discrepancy. For distributional fidelity, we use kernel Stein discrepancy (KSD) with score $s_p(x) = \nabla_x \log p(x)$ and an RBF base kernel $k_h(\cdot,\cdot)$ [13].

The corresponding Stein kernel is

$$\begin{aligned}\kappa_p(x,x') = s_p(x)^\top s_p(x')\, k_h(x,x') + s_p(x)^\top \nabla_{x'} k_h(x,x') \\ + s_p(x')^\top \nabla_x k_h(x,x') + \mathrm{tr}(\nabla_x \nabla_{x'} k_h(x,x'))\,,\end{aligned} \tag{15}$$

and the empirical discrepancy is

$$\mathrm{KSD}(X;p) = \left(\frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \kappa_p(x_i, x_j) \right)^{1/2}. \tag{16}$$

Under standard kernel conditions, KSD is zero only at the target law and increases under global mismatch, including location bias, covariance distortion, and mode omission [13].

Mean log-density fit. We complement KSD with the empirical mean log-density

$$\overline{\log p}(X) = \frac{1}{N} \sum_{i=1}^{N} \log p(x_i), \tag{17}$$

which corresponds to the logarithmic scoring rule [14]. This metric quantifies average concentration in high-density regions, but it is local in $\log p(x)$ and therefore does not, by itself, certify global distributional agreement. Reporting eqs. (16) and (17) separates global fit from local density concentration.

Predictive performance and calibration. For UCI and BNN tasks, we report accuracy, negative log-likelihood $\mathrm{NLL} = -\frac{1}{M} \sum_{m=1}^{M} \big[y_m \log \hat{p}_m + (1 - y_m) \log(1 - \hat{p}_m) \big]$, and expected calibration error $\mathrm{ECE} = \sum_{b=1}^{B} \frac{|\mathcal{B}_b|}{M} \, |\mathrm{acc}(\mathcal{B}_b) - \mathrm{conf}(\mathcal{B}_b)|$. This combination separates discrimination (accuracy), probabilistic sharpness (NLL), and calibration quality (ECE).

Mode-structure diagnostics. For mixture and grid-based 2D targets, we report mode coverage, normalized mode entropy, minimum mode mass, and grid L_1 discrepancy to detect mode collapse and quantify spatial mismatch with reference densities.

Cost accounting. In addition to wall-clock time, we track gradient- and kernel-evaluation counts to support hardware-agnostic efficiency comparisons across methods with different per-step workloads.

Mixing efficiency. To quantify dependence, we report ESS from a one-dimensional diagnostic sequence $z_1, \dots, z_T$ built from the first coordinate in our implementation. For particle methods at a checkpoint, z_i is the first coordinate of particle i ($i = 1, \dots, N$). For single-chain baselines, z_i is the first coordinate of the i-th state in a rolling window of recent iterates (window length as set in each benchmark script). Define

$$\bar{z} = \frac{1}{T}\sum_{i=1}^{T} z_i, \qquad \hat{\gamma}_k = \frac{1}{T-k}\sum_{i=1}^{T-k}(z_i - \bar{z})(z_{i+k} - \bar{z}), \qquad \hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}. \tag{18}$$

Autocorrelations are evaluated up to $k_{\max} = \min(\lfloor T/2 \rfloor, 100)$. Let K be the first lag in $\{1, \dots, k_{\max}\}$ such that $\hat{\rho}_K < 0$; if no such lag exists, we set $K = k_{\max}$. The reported ESS is

$$\mathrm{ESS}(z) = \frac{T}{1 + 2\sum_{k=1}^{K} \hat{\rho}_k}. \tag{19}$$

This initial-positive truncation controls high-lag noise in the empirical autocorrelation and provides a stable scalar proxy for mixing. Because eq. (19) is computed from a single diagnostic coordinate, we interpret ESS as a univariate mixing indicator rather than a full multivariate efficiency estimate. Accordingly, distribution-level conclusions rely primarily on KSD and moment/mode diagnostics, with ESS used as a complementary statistic.

Lower is better for e_μ, e_Σ, KSD, NLL, ECE, and grid L_1, whereas higher is better for ESS, accuracy, coverage, entropy, minimum mode mass, and mean log-probability.

3.2. Experimental protocol

Across benchmarks, we compare the same method family whenever applicable: vanilla SVGD, Strang-split SVGD, fixed multirate SVGD (MR-SVGD), adaptive multirate SVGD (Adapt-MR-SVGD), and the single-chain baselines SGLD and an SGHMC-style method. The SVGD-family updates are those defined in section 2, namely eqs. (1), (7), (8) and (14).

Remark 2. For the single-chain baselines, we use the standard stochastic-gradient update

$$x_{t+1} = x_t + \eta \nabla \log p(x_t) + \sqrt{2\eta}\, \xi_t, \qquad \xi_t \sim \mathcal{N}(0, I), \tag{20}$$

for SGLD [1], and

$$v_{t+1} = \beta v_t + \eta \nabla \log p(x_t) + \sqrt{2\alpha\eta}\, \xi_t, \qquad x_{t+1} = x_t + v_{t+1}, \tag{21}$$

for an SGHMC-style baseline [2]. Here η is the step size, α is a friction/noise parameter, and $\beta \in (0, 1)$ is momentum decay.

For particle methods, we use $N = 128$ particles; for single-chain methods, we compute metrics from a rolling window of recent iterates to obtain comparable finite-sample summaries. Alongside wall-clock time, we report gradient- and kernel-evaluation counts as hardware-agnostic measures of computational cost.

All benchmarks use checkpointed evaluation at interval `save_every` and an early-stopping rule triggered by persistent degradation of a designated monitor metric. We use KSD as the monitor for 50D Gaussian, 2D synthetic targets, Mixture2D, and BNN (`COMPUTE_KSD=True` in the BNN setup), whereas UCI logistic regression uses NLL as the monitor because its default configuration sets `COMPUTE_KSD=False`.

3.3. 50D Gaussian

This benchmark stresses stability in a high-dimensional setting with anisotropic covariance structure. We run 1,000 iterations with checkpointing every 50 steps and a 5-seed ablation. Reported metrics are moment errors ($\|\hat{\mu} - \mu\|_2$ and $\|\hat{\Sigma} - \Sigma\|_F$), ESS, KSD, and mean log probability.

3.4. 2D synthetic targets

We evaluate banana, ring, squiggly, two-moons, and funnel targets with 1,000 iterations and checkpointing every 20 steps. Metrics include mean error, covariance error, ESS, KSD, mean log probability, and grid-based L_1 discrepancy against reference densities. The current draft keeps a placeholder for a consolidated multi-target figure; per-target panels are available under `figures/2d/<target>/`.

3.5. Mixture2D (mix8)

This benchmark emphasizes mode discovery and mass allocation across modes. We run 1,000 iterations (checkpoint every 20) across 5 seeds. Metrics include mode coverage, mode entropy, imbalance, minimum mass per mode, ESS, KSD, mean log probability, and grid L_1 discrepancy. The summary panels directly visualize coverage/entropy-quality tradeoffs together with cost-aware curves.

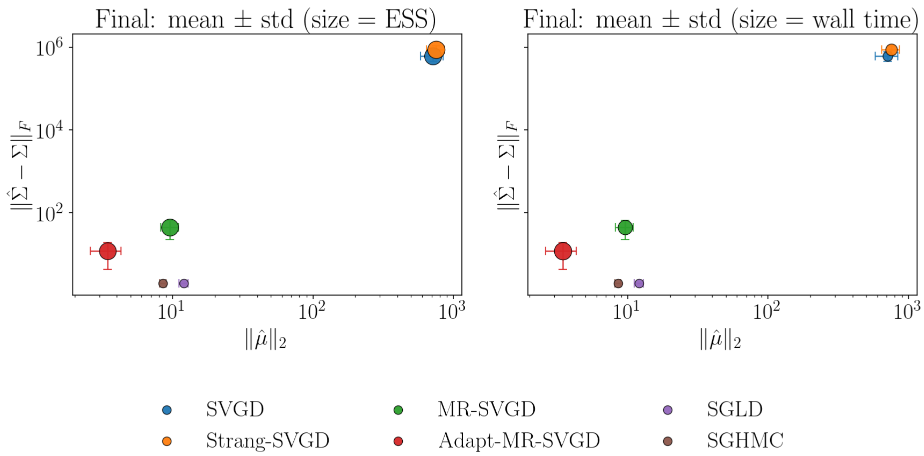

Figure 1: 50D Gaussian: final mean ± std Pareto plot in moment-error space with marker size encoding ESS (left) and wall time (right).

Method	Med. L_1 ↓	Med. KSD↓	Med. grad evals↓	Med. wall (s)↓	Pareto
SVGD	1.964	604.15	180	0.77	Yes
Strang-SVGD	1.997	57.17	220	2.44	Yes
MR-SVGD	1.999	68.27	180	3.11	No
Adapt-MR-SVGD	1.999	1.94	1004	2.89	Yes

Table 1: 2D targets quality-cost summary across particle methods. Entries are medians across targets of final-checkpoint metrics from current outputs (`metrics/2d/*.csv`). Lower is better for L_1 and KSD. The Pareto flag is computed on the (median KSD, median wall-time) plane.

3.6. UCI logistic regression

We benchmark Bayesian logistic regression on breast_cancer, ionosphere, spambase, and a5a, using a Gaussian prior on weights (as implemented in the benchmark script) and 5 seeds per dataset. We report test accuracy, NLL, ECE, ESS, and mean log probability; KSD is optional and disabled by default in this benchmark configuration. Results are summarized with per-dataset panels in fig. 6 and a cross-dataset winner scorecard (particle methods only) in table 2 [15].

3.7. Bayesian neural network

We evaluate a one-hidden-layer BNN (hidden width 32) on breast_cancer, ionosphere, and a5a, using a Gaussian prior and 5 seeds per dataset. We report test accuracy, NLL, ECE, ESS, KSD, and mean log probability, and

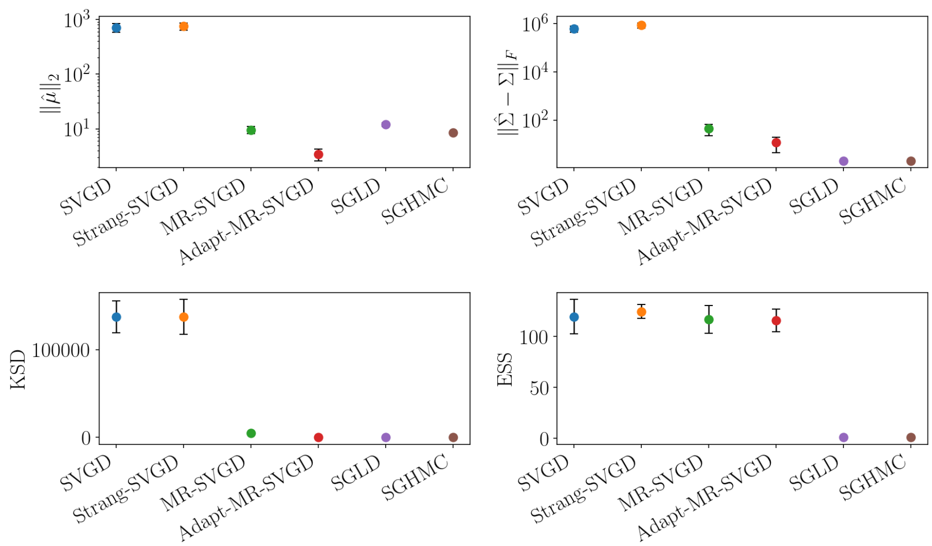

Figure 2: 50D Gaussian final-checkpoint summary across methods. The four panels report $\|\hat{\mu} - \mu\|_2$, $\|\hat{\Sigma} - \Sigma\|_F$, KSD, and ESS, respectively; markers show mean values across seeds and error bars indicate one standard deviation. Lower is better for the first three metrics, whereas higher is better for ESS.

summarize results with panels analogous to the logistic-regression benchmark.

3.8. Large-scale hierarchical logistic regression

To stress-test multirate behavior in a high-dimensional and strongly anisotropic posterior, we use a large-scale hierarchical logistic regression benchmark with grouped random effects. This setting combines substantial parameter dimension with funnel-like geometry induced by a global scale parameter, making it a natural target for timing-quality comparisons among SVGD variants.

Problem definition. Given binary outcomes $y_i \in \{0, 1\}$, sparse features $x_i \in \mathbb{R}^p$, and group indices $g_i \in \{1, \ldots, G\}$, we define

$$y_i \mid x_i, g_i, \beta, \alpha, u \sim \mathrm{Bernoulli}\big(\sigma\big(\alpha + x_i^\top \beta + u_{g_i}\big)\big)\,, \qquad i = 1, \ldots, n, \tag{22}$$

where $\sigma(\cdot)$ is the logistic link. We use a non-centered parameterization for group effects,

$$\beta \sim \mathcal{N}(0, \sigma_\beta^2 I_p), \quad \alpha \sim \mathcal{N}(0, \sigma_\alpha^2), \quad u_j = \tau z_j, \quad z_j \sim \mathcal{N}(0, 1), \quad \log \tau \sim \mathcal{N}(\mu_\tau, \sigma_\tau^2), \tag{23}$$

Dataset	Acc↑ winner	NLL↓ winner	ECE↓ winner	ESS↑ winner
breast_cancer	MR (0.977)	Adapt-MR (0.075)	SVGD (0.594)	MR (124.0)
ionosphere	SVGD (0.891)	Adapt-MR (0.523)	SVGD (0.378)	MR (127.7)
spambase	SVGD (0.884)	Adapt-MR (0.412)	Adapt-MR (0.541)	SVGD/MR (128.0)
a5a	Strang (0.815)	MR (0.510)	Adapt-MR (0.612)	SVGD (123.2)

Table 2: UCI logistic regression winner scorecard (particle methods only: SVGD, Strang-SVGD, MR-SVGD, Adapt-MR-SVGD). Each entry reports the best method and its final-checkpoint mean value across seeds for the corresponding dataset and metric; ties are shown with “/”.

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.613	0.658	123.3
MR-SVGD	5/5	0.623	0.656	92.5
SGHMC	2/5	0.860	0.560	77.9
SVGD	3/5	0.885	0.577	154.2
Strang-SVGD	3/5	1.263	0.579	163.6
SGLD	4/5	1.823	0.576	101.6

Table 3: HLR long-tail summary over five seeds. For each seed and method, we select the best finite-NLL checkpoint; the table reports method-wise means of that best NLL, the accuracy at that checkpoint, and the corresponding wall time. “Seeds with finite best” counts runs that reached at least one finite-NLL checkpoint before stopping.

for $j = 1, \dots, G$. The posterior (up to normalization) is

$$\pi(\theta) \propto \left[\prod_{i=1}^{n} \mathrm{Bernoulli}\big(y_i; \sigma\big(\alpha + x_i^\top \beta + u_{g_i}\big)\big)\right] p(\beta)\, p(\alpha)\, p(z)\, p(\log \tau), \qquad \theta = (\beta, \alpha, z, \log \tau). \tag{24}$$

Protocol and stopping rule. For the long-tail grouped setting, we run $n = 10^6$ observations with $p = 300$ features and $G = 50{,}000$ groups. Each method runs up to 1,000 iterations with checkpointing every 20 iterations and 5 random seeds; particle methods use 32 particles, and single-chain baselines use a rolling-window evaluation set. We use NLL-only early stopping with patience and a non-finite fail-fast rule; ECE is reported as a metric but is not used as a stopping trigger.

Result summary. table 3 shows that Adapt-MR-SVGD achieves the best mean best-NLL while remaining finite across all seeds. MR-SVGD is a close second in NLL and reaches its best checkpoint faster on average (92.5 s versus 123.3 s). SVGD/Strang-SVGD and SGHMC show frequent non-finite

Method	Finite-best↑	Best NLL↓	Acc@best NLL↑	Wall@best (s)↓
Adapt-MR-SVGD	5/5	0.681	0.601	94.6
SVGD	1/5	0.700	0.548	54.9
MR-SVGD	5/5	0.706	0.597	44.2
Strang-SVGD	1/5	0.708	0.584	241.7
SGLD	5/5	0.743	0.528	66.0
SGHMC	5/5	0.791	0.554	79.7

Table 4: HLR uniform-group summary over five seeds. As in table 3, each entry uses the best finite-NLL checkpoint per seed and reports method-wise means for NLL, accuracy at that checkpoint, and wall time.

behavior in this regime, while SGLD remains finite more often but with substantially worse NLL.

Uniform-group sensitivity check. table 4 provides a compact robustness check under uniform group assignments. Adapt-MR-SVGD remains best in mean best-NLL with full finite-seed coverage (5/5), and MR-SVGD remains close while being fastest among methods with full finite coverage. This supports the same ordering seen in the long-tail regime while keeping the long-tail setting as the primary stress test for our main claim.

4. Conclusions

This work reframes SVGD as a numerically split interacting-particle flow in which attraction to high-density regions and diversity-preserving repulsion are treated as distinct computational components. From this viewpoint, multirate integration is not only an efficiency heuristic but a practical mechanism for improving stability in posterior geometries where these components evolve at different effective scales.

Across a broad benchmark suite, the main empirical pattern is that multirate formulations make SVGD behavior more reliable under difficult geometry, especially in high-dimensional anisotropic settings and hierarchical models. Fixed multirate schedules provide a strong robustness baseline, while adaptive drift resolution offers additional flexibility when local stiffness changes along the trajectory. At the same time, results indicate that no single variant dominates every metric and every regime, reinforcing the importance of evaluating quality and cost jointly rather than through a single summary statistic.

Several limitations remain. Current adaptivity is centered on drift resolution, while repulsion frequency is still controlled by simple schedules; ESS remains a univariate proxy; and dense kernel interactions continue to constrain scaling at large particle counts. These observations suggest clear next steps: tighter adaptive controllers that coordinate drift and repulsion updates, approximate or structured kernel computations for large-N regimes, and stronger theoretical links between multirate stability analysis and particle-based Bayesian inference guarantees. Together, these directions can move multirate SVGD from a robust empirical strategy toward a more principled and scalable foundation for modern Bayesian computation.

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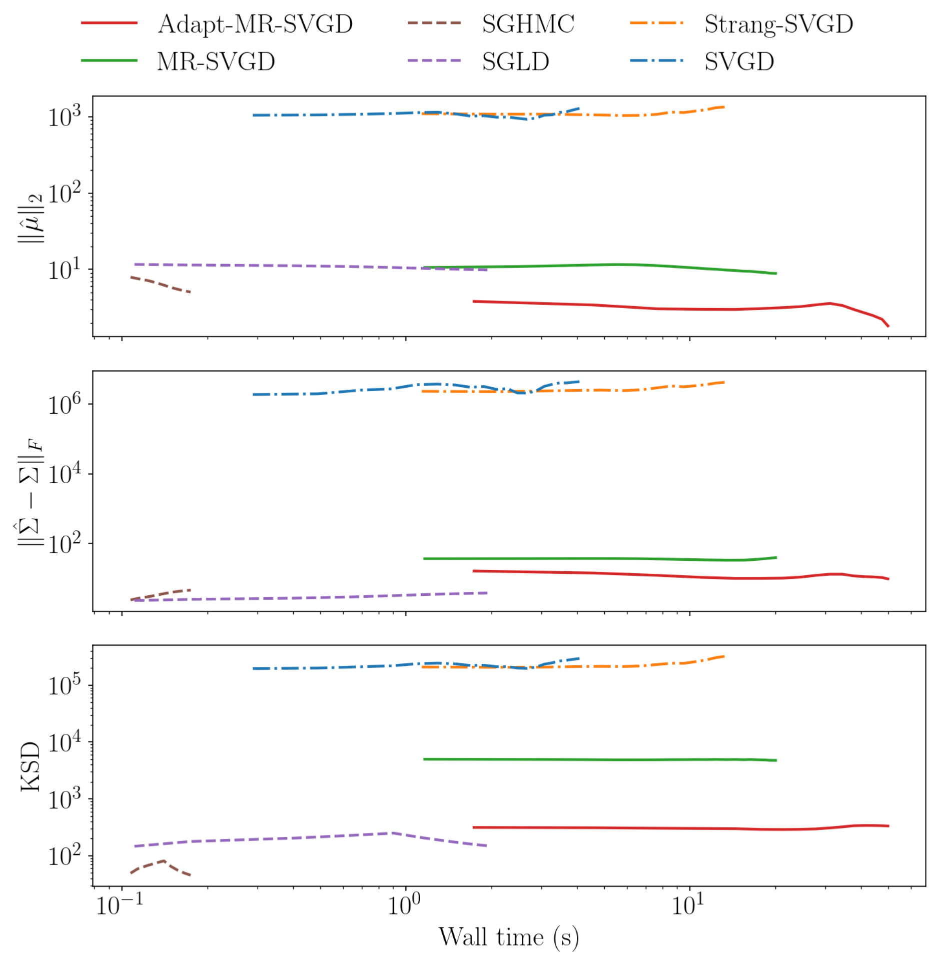


Figure 3: 50D Gaussian: wall-time overview across key metrics.

Placeholder for 2D summary figure.

Figure 4: 2D targets: placeholder for the main figure.

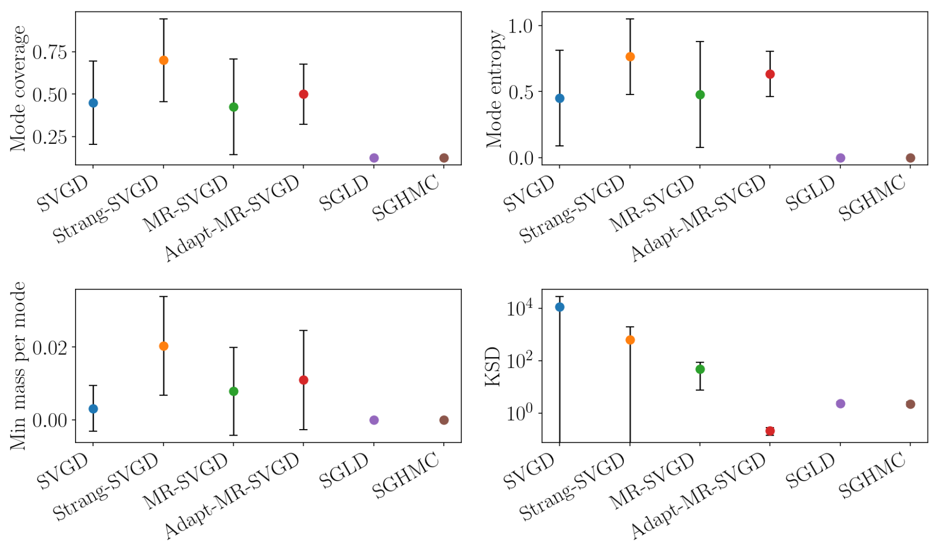
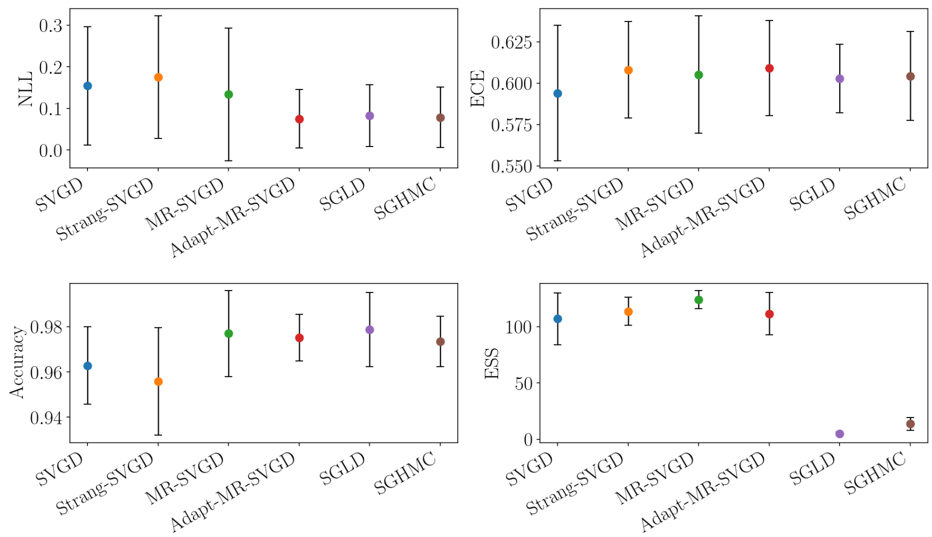
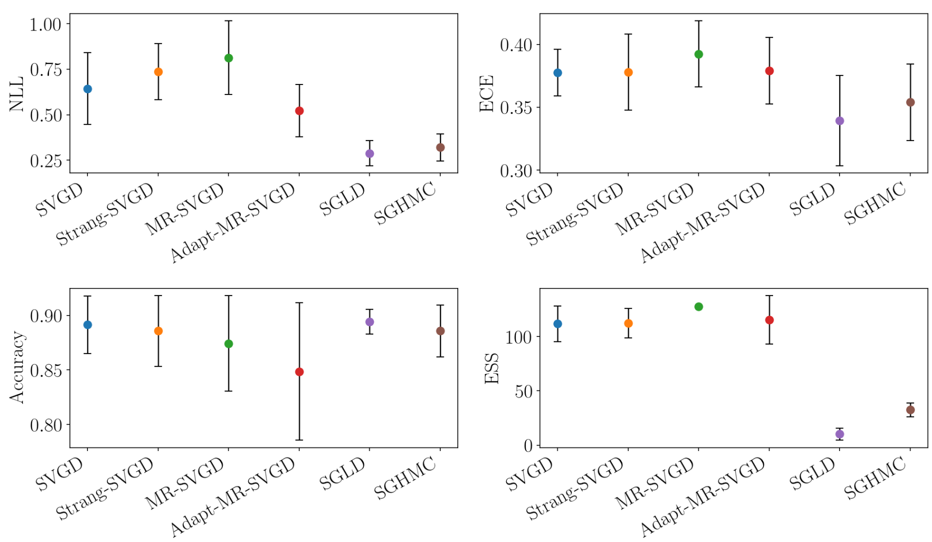


Figure 5: Mixture2D (mix8) final-checkpoint summary across methods. The four panels report mode coverage, mode entropy, minimum mass per mode, and KSD; markers show mean values across seeds and error bars indicate one standard deviation. Higher is better for coverage, entropy, and minimum mass per mode, while lower is better for KSD.

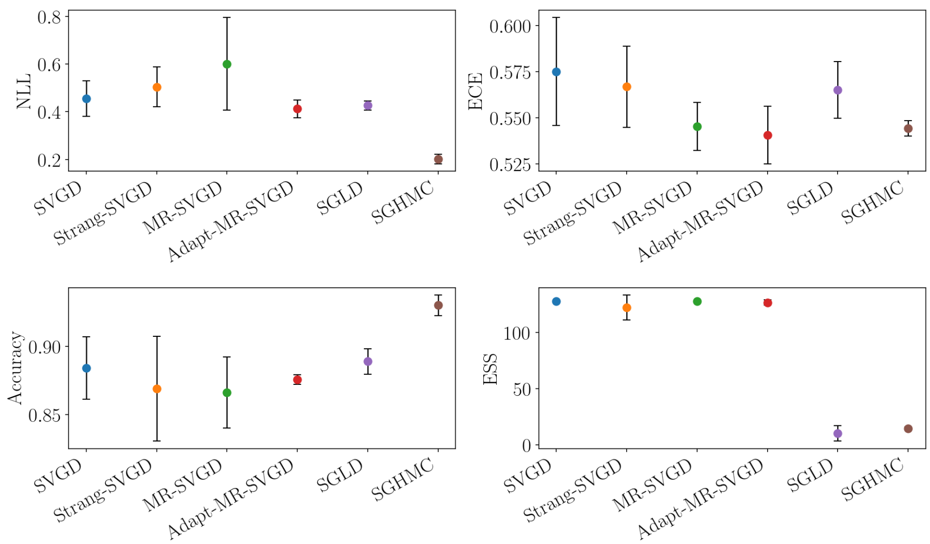


(a) breast_cancer

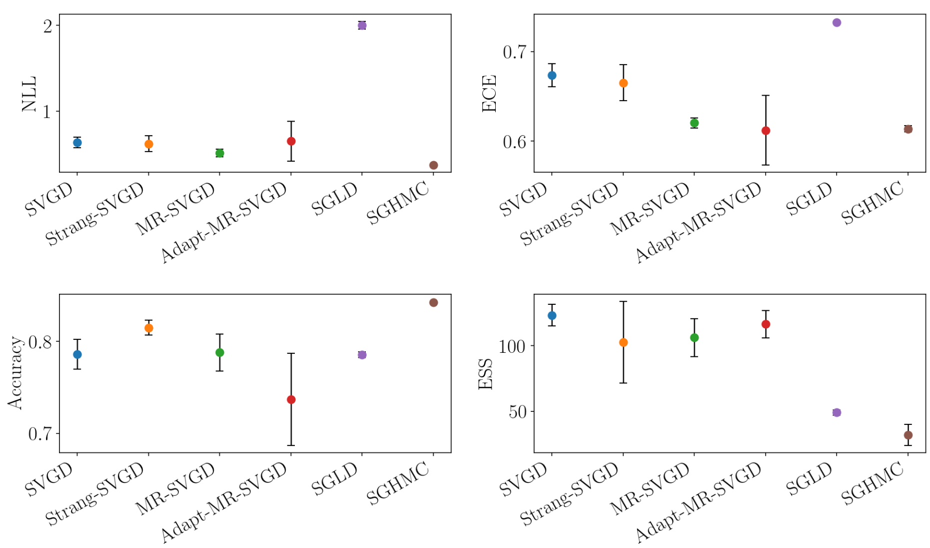


(b) ionosphere

Figure 6: UCI logistic regression summary across datasets. Each subpanel reports final-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS. Higher is better for accuracy and ESS; lower is better for NLL and ECE. Remaining datasets are shown in the continued panel.

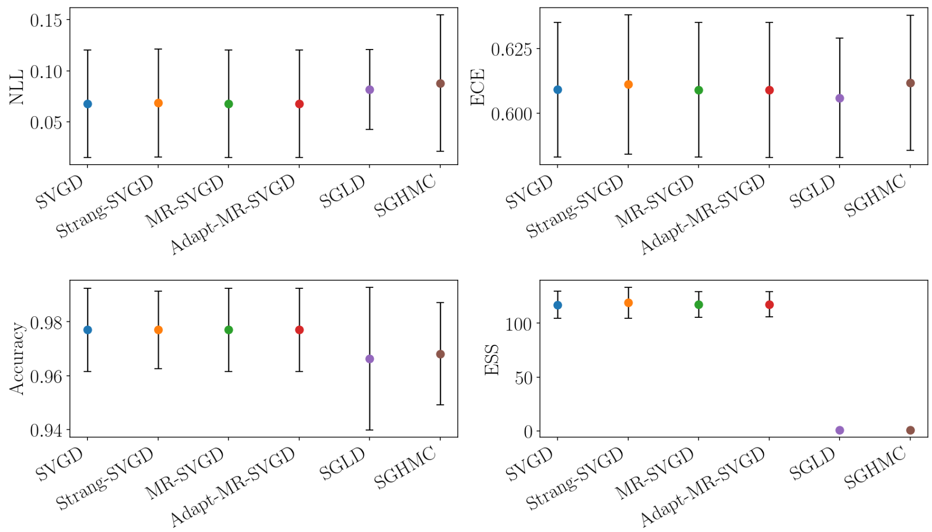


(c) spambase

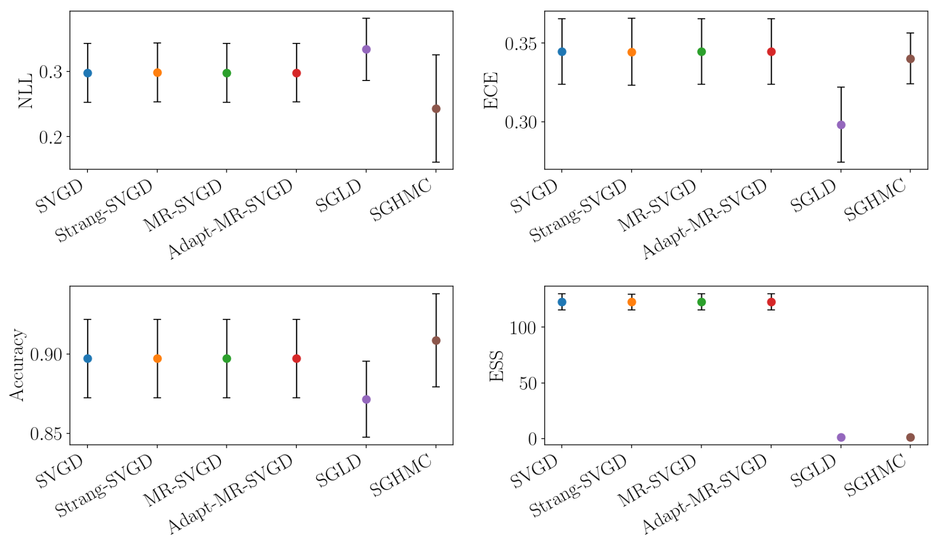


(d) a5a

Figure 6: UCI logistic regression summary across datasets (continued).

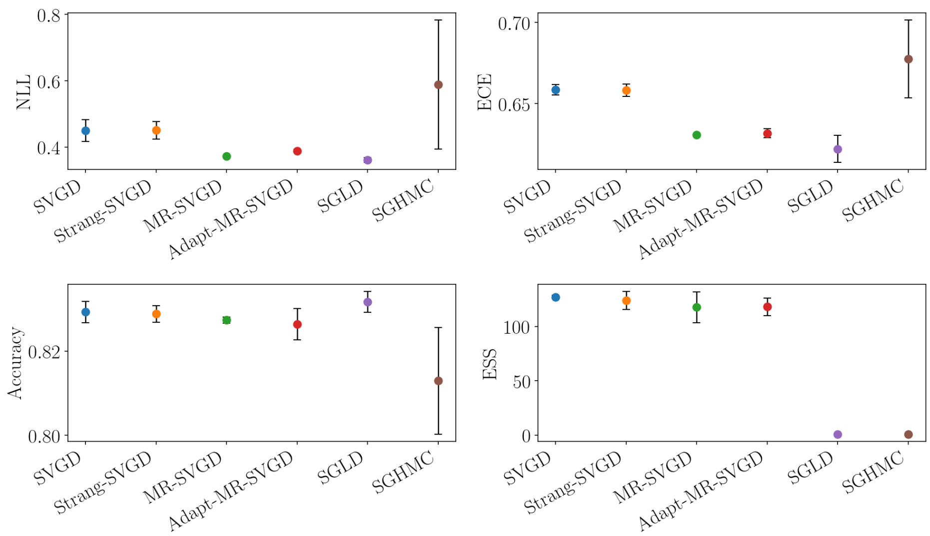


(a) breast_cancer



(b) ionosphere

Figure 7: BNN summary across datasets. Each subpanel reports final-checkpoint mean across seeds with one standard deviation for test accuracy, NLL, ECE, and ESS. Higher is better for accuracy and ESS; lower is better for NLL and ECE. The remaining dataset appears in the continued panel.



(c) a5a

Figure 7: BNN summary across datasets (continued).